

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

```

```

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(GP_FFCutsFits);
for i = 1:numel(fn)
    selectCuts = fn{i};
    if contains(selectCuts, 'msk')
        continue
    end
    if contains(selectCuts, 'all_fit_par')
        continue
    end

    fprintf('Processing cuts: %s \n', selectCuts);
    %selectCuts= 'Ei400_off200GPmaxFF';
    cuts_list = GP_FFCutsFits.(selectCuts);
    if ~iscell(cuts_list)
        cuts_list = {cuts_list};
    end
    if contains(selectCuts, 'Ei200')
        mi = maps_instrument(200,600, 'S');
    end
    if contains(selectCuts, 'Ei400')
        mi = maps_instrument(401,600, 'S');
    end
    if contains(selectCuts, 'Ei800')
        mi = maps_instrument(786,600, 'S');
    end
    sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
    for j=1:numel(cuts_list)
        cuts_list{j} = cuts_list{i}.set_instrument(mi);
        cuts_list{j} = cuts_list{i}.set_sample(sample);
        plot(cuts_list{j}); lz 0 1; keep_figure;
    end
    dir_name = "GP";

    dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
    half_dE = 5; % half width of data binning

    en_mm = cuts_list{1}.data.img_range(:,4);

```

```

en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
cl = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep

all_fit_par = GP_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

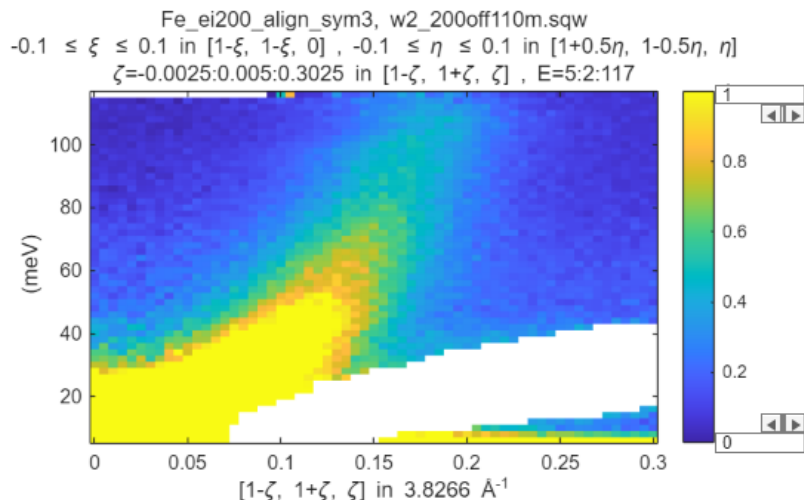
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;

%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure

doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
close all;
end

```

Processing cuts: Ei200_off110Gpmiff



**** En = 35±5 going to 105

Beginning fit (max 20 iterations)-----

Starting point

 Total time = 12.508s Reduced Chi^2 = 368.3939

Free parameter values:

10 0.7917 33.5 0.862

 Iteration = 1

 Total time = 29.7708s Reduced Chi^2 = 96.0201 Levenberg-Marquardt = 0.1

Free parameter values:

11.81 0.8392 19.52 0.3982

 Iteration = 2

 Total time = 42.5175s Reduced Chi^2 = 15.5677 Levenberg-Marquardt = 0.01

Free parameter values:

19.94 1.34 26.03 0.2415

 Iteration = 3

 Total time = 54.8013s Reduced Chi^2 = 128.5389 Levenberg-Marquardt = 0.001

Free parameter values:

-6.545 0.3209 19.99 0.3587

Total time = 57.9059s Reduced Chi^2 = 46.4982 Levenberg-Marquardt = 0.01

Free parameter values:

0.1745 0.6575 21.73 0.3085

Total time = 60.9658s Reduced Chi^2 = 4.631 Levenberg-Marquardt = 0.1

Free parameter values:

9.573 1.109 24.19 0.2464

 Iteration = 4

 Total time = 73.3703s Reduced Chi^2 = 2.8896 Levenberg-Marquardt = 0.01

Free parameter values:

6.916 0.8851 24.04 0.3182

 Iteration = 5

 Total time = 85.6363s Reduced Chi^2 = 2.7478 Levenberg-Marquardt = 0.001

Free parameter values:

6.413 0.8745 24.11 0.3261

```

-----
Iteration = 6
-----
Total time = 98.1307s    Reduced Chi^2 = 2.7271    Levenberg-Marquardt = 0.0001
Free parameter values:
    6.33      0.8696      24.12      0.3274
-----
Iteration = 7
-----
Total time = 110.5301s    Reduced Chi^2 = 2.7931    Levenberg-Marquardt = 1e-05
Free parameter values:
    6.374      0.8713      24.13      0.3272
Total time = 113.6309s    Reduced Chi^2 = 2.6824    Levenberg-Marquardt = 0.0001
Free parameter values:
    6.374      0.8713      24.13      0.3272
-----
Iteration = 8
-----
Total time = 126.0343s    Reduced Chi^2 = 2.6961    Levenberg-Marquardt = 1e-05
Free parameter values:
    6.328      0.8701      24.13      0.3278
Total time = 129.1201s    Reduced Chi^2 = 2.7344    Levenberg-Marquardt = 0.0001
Free parameter values:
    6.328      0.8701      24.13      0.3278
Total time = 132.2065s    Reduced Chi^2 = 2.7564    Levenberg-Marquardt = 0.001
Free parameter values:
    6.329      0.8701      24.13      0.3278
Total time = 135.2828s    Reduced Chi^2 = 2.7879    Levenberg-Marquardt = 0.1
Free parameter values:
    6.342      0.8708      24.13      0.3276
Total time = 138.4005s    Reduced Chi^2 = 2.7201    Levenberg-Marquardt = 10
Free parameter values:
    6.366      0.8714      24.13      0.3273
Total time = 141.4689s    Reduced Chi^2 = 2.7013    Levenberg-Marquardt = 1000
Free parameter values:
    6.373      0.8713      24.13      0.3272
*** No improvement in chi-squared over previous iteration ***
Total time = 141.4765s    Reduced Chi^2 = 2.6824
Free parameter values:
    6.374      0.8713      24.13      0.3272
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3          6.374 +/- 0.1302
4          0.8713 +/- 0.01704
5          0
6          24.13 +/- 0.1511
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1          0.3272 +/- 0.0104
2          0

```

Covariance matrix for free parameters:

```

1.0000    0.3836   -0.4691   -0.7866
0.3836    1.0000   -0.0633   -0.8494
-0.4691   -0.0633    1.0000    0.1443
-0.7866   -0.8494    0.1443    1.0000

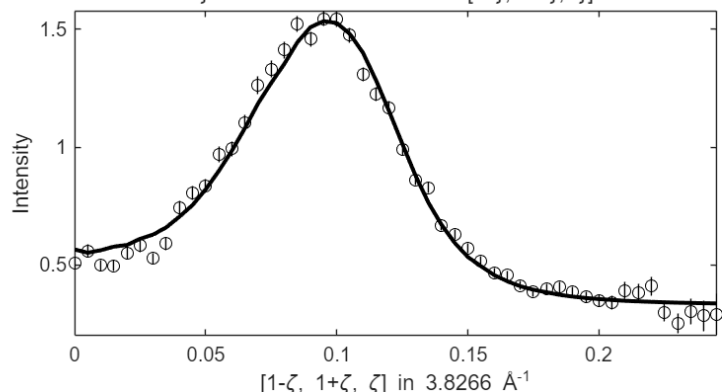
```

Fe_ei200_align_sym3, w2_200off110m.sqw

S=0.87±0.017; J0= 24±0.15; gamma=6.37426±0.130249

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 30 ≤ E

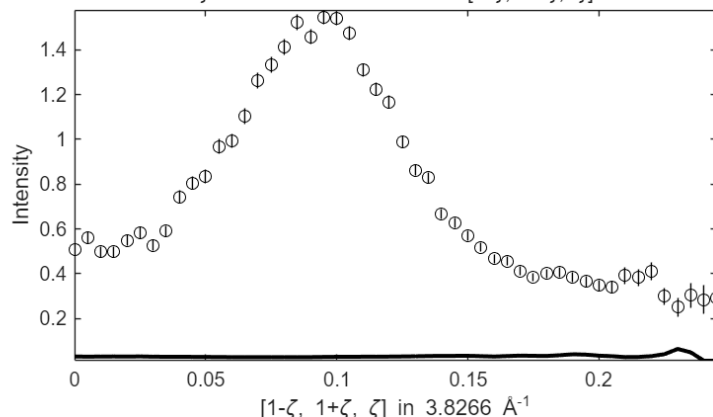
ζ=-0.0025:0.005:0.3025 in [1-ζ, 1+ζ, ζ]



Fe_ei200_align_sym3, w2_200off110m.sqw

.1 ≤ ξ ≤ 0.1 in [1-ξ, 1-ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1+0.5η, 1-0.5η, η] , 30 ≤ E

ζ=-0.0025:0.005:0.3025 in [1-ζ, 1+ζ, ζ]



**** En = 45±5 going to 105

Beginning fit (max 20 iterations)-----

Starting point

Total time = 6.3541s Reduced Chi^2 = 418.0962

Free parameter values:

6.374 0.8713 24.13 0.5316

Iteration = 1

Total time = 26.7502s Reduced Chi^2 = 3.1706 Levenberg-Marquardt = 0.1

Free parameter values:

7.871 0.7877 24.39 0.2332

Iteration = 2

```

Total time = 46.2646s    Reduced Chi^2 = 1.6443    Levenberg-Marquardt = 0.01
Free parameter values:
    10.03      0.8744      24.56      0.2022
-----
Iteration = 3
-----
Total time = 65.5683s    Reduced Chi^2 = 1.6269    Levenberg-Marquardt = 0.001
Free parameter values:
    10.27      0.8872      24.57      0.2003
-----
Iteration = 4
-----
Total time = 84.8962s    Reduced Chi^2 = 1.6375    Levenberg-Marquardt = 0.0001
Free parameter values:
    10.28      0.8865      24.57      0.2002
Total time = 89.8747s    Reduced Chi^2 = 1.6492    Levenberg-Marquardt = 0.001
Free parameter values:
    10.28      0.8865      24.57      0.2002
Total time = 94.9058s    Reduced Chi^2 = 1.6347    Levenberg-Marquardt = 0.01
Free parameter values:
    10.28      0.8865      24.57      0.2002
Total time = 99.7674s    Reduced Chi^2 = 1.6524    Levenberg-Marquardt = 1
Free parameter values:
    10.28      0.8868      24.57      0.2002
Total time = 104.652s    Reduced Chi^2 = 1.6358    Levenberg-Marquardt = 100
Free parameter values:
    10.27      0.8872      24.57      0.2002
Total time = 109.5906s    Reduced Chi^2 = 1.6427    Levenberg-Marquardt = 10000
Free parameter values:
    10.27      0.8872      24.57      0.2003
*** No improvement in chi-squared over previous iteration ***
Total time = 109.5982s    Reduced Chi^2 = 1.6269
Free parameter values:
    10.27      0.8872      24.57      0.2003
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
    1          1
    2          8
    3      10.27 +/- 0.1413
    4      0.8872 +/- 0.01083
    5          0
    6      24.57 +/- 0.1039
    7          0
    8          0
    9          0
   10          0

Background parameter values:
-----
    1      0.2003 +/- 0.004034
    2          0
    3          0

Covariance matrix for free parameters:
-----
    1.0000    0.3071   -0.3939   -0.7870

```

0.3071	1.0000	-0.0718	-0.7991
-0.3939	-0.0718	1.0000	0.1023
-0.7870	-0.7991	0.1023	1.0000

 **** En = 55±5 going to 105

 Beginning fit (max 20 iterations)-----

Starting point

 Total time = 5.8799s Reduced Chi^2 = 474.7294

Free parameter values:

10.27	0.8872	24.57	0.4171
-------	--------	-------	--------

 Iteration = 1

 Total time = 28.9132s Reduced Chi^2 = 4.0396 Levenberg-Marquardt = 0.1

Free parameter values:

12.43	0.7679	25.25	0.1694
-------	--------	-------	--------

 Iteration = 2

 Total time = 51.5889s Reduced Chi^2 = 2.3379 Levenberg-Marquardt = 0.01

Free parameter values:

15.73	0.8709	25.73	0.1405
-------	--------	-------	--------

 Iteration = 3

 Total time = 74.3801s Reduced Chi^2 = 2.2948 Levenberg-Marquardt = 0.001

Free parameter values:

16.22	0.8909	25.78	0.1381
-------	--------	-------	--------

 Iteration = 4

 Total time = 97.4169s Reduced Chi^2 = 2.3044 Levenberg-Marquardt = 0.0001

Free parameter values:

16.2	0.8902	25.78	0.1382
------	--------	-------	--------

Total time = 103.2791s Reduced Chi^2 = 2.2995 Levenberg-Marquardt = 0.001

Free parameter values:

16.2	0.8903	25.78	0.1382
------	--------	-------	--------

Total time = 109.1592s Reduced Chi^2 = 2.3037 Levenberg-Marquardt = 0.01

Free parameter values:

16.2	0.8903	25.78	0.1382
------	--------	-------	--------

Total time = 114.9508s Reduced Chi^2 = 2.3116 Levenberg-Marquardt = 1

Free parameter values:

16.21	0.8908	25.78	0.1381
-------	--------	-------	--------

Total time = 120.723s Reduced Chi^2 = 2.3012 Levenberg-Marquardt = 100

Free parameter values:

16.22	0.8909	25.78	0.1381
-------	--------	-------	--------

Total time = 126.5062s Reduced Chi^2 = 2.3162 Levenberg-Marquardt = 10000

Free parameter values:

16.22	0.8909	25.78	0.1381
-------	--------	-------	--------

*** No improvement in chi-squared over previous iteration ***

Total time = 126.5167s Reduced Chi^2 = 2.2948

Free parameter values:

16.22	0.8909	25.78	0.1381
-------	--------	-------	--------

 Fit converged:

Parameter values (free parameters with error estimates):

 Foreground parameter values:

```

-----
1          1
2          8
3          16.22 +/- 0.3125
4          0.8909 +/- 0.01647
5          0
6          25.78 +/- 0.1236
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1          0.1381 +/- 0.004628
2          0
3          0

```

Covariance matrix for free parameters:

```

-----
1.0000    0.5953   -0.1927   -0.8741
0.5953    1.0000   -0.0112   -0.8944
-0.1927   -0.0112    1.0000   -0.0253
-0.8741   -0.8944   -0.0253    1.0000

```

```

*****
**** En = 65±5 going to 105
*****

```

Beginning fit (max 20 iterations)-----

Starting point

```

-----
Total time = 6.8697s    Reduced Chi^2 = 496.7905
Free parameter values:
    16.22      0.8909      25.78      0.3586

```

Iteration = 1

```

-----
Total time = 32.8046s    Reduced Chi^2 = 3.1031    Levenberg-Marquardt = 0.1
Free parameter values:
    16.32      0.7714      26.62      0.1379

```

Iteration = 2

```

-----
Total time = 58.6824s    Reduced Chi^2 = 1.2748    Levenberg-Marquardt = 0.01
Free parameter values:
    19.53      0.884      27.07      0.1095

```

Iteration = 3

```

-----
Total time = 84.4508s    Reduced Chi^2 = 1.2446    Levenberg-Marquardt = 0.001
Free parameter values:
    19.98      0.9028      27.13      0.1073

```

Iteration = 4

```

-----
Total time = 110.4114s    Reduced Chi^2 = 1.2575    Levenberg-Marquardt = 0.0001
Free parameter values:
    19.92      0.9013      27.12      0.1075
Total time = 116.9243s    Reduced Chi^2 = 1.2592    Levenberg-Marquardt = 0.001
Free parameter values:
    19.92      0.9013      27.12      0.1074

```

```

Total time = 123.4671s   Reduced Chi^2 = 1.2517   Levenberg-Marquardt = 0.01
Free parameter values:
    19.93      0.9015      27.12      0.1074
Total time = 129.8963s   Reduced Chi^2 = 1.263     Levenberg-Marquardt = 1
Free parameter values:
    19.96      0.9025      27.12      0.1073
Total time = 136.4132s   Reduced Chi^2 = 1.2526     Levenberg-Marquardt = 100
Free parameter values:
    19.98      0.9027      27.13      0.1073
Total time = 142.9768s   Reduced Chi^2 = 1.2472     Levenberg-Marquardt = 10000
Free parameter values:
    19.98      0.9028      27.13      0.1073
*** No improvement in chi-squared over previous iteration ***
Total time = 142.9846s   Reduced Chi^2 = 1.2446
Free parameter values:
    19.98      0.9028      27.13      0.1073

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	19.98	+/-	0.332
4	0.9028	+/-	0.01433
5	0		
6	27.13	+/-	0.0918
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.1073	+/-	0.003293
2	0		
3	0		

Covariance matrix for free parameters:

1.0000	0.7449	-0.0078	-0.9248
0.7449	1.0000	0.1093	-0.9317
-0.0078	0.1093	1.0000	-0.1633
-0.9248	-0.9317	-0.1633	1.0000

**** En = 75±5 going to 105

Beginning fit (max 20 iterations)-----

Starting point

```

Total time = 7.5003s   Reduced Chi^2 = 582.6575
Free parameter values:
    19.98      0.9028      27.13      0.2977

```

Iteration = 1

```

Total time = 37.1524s   Reduced Chi^2 = 3.2567   Levenberg-Marquardt = 0.1

```

```

Free parameter values:
    19.22      0.7358      27.69      0.1008
-----
Iteration = 2
-----
Total time = 66.6761s    Reduced Chi^2 = 1.3936    Levenberg-Marquardt = 0.01
Free parameter values:
    22.51      0.8397      28.07      0.07615
-----
Iteration = 3
-----
Total time = 96.7302s    Reduced Chi^2 = 1.36    Levenberg-Marquardt = 0.001
Free parameter values:
    22.97      0.8564      28.12      0.07422
-----
Iteration = 4
-----
Total time = 126.3487s    Reduced Chi^2 = 1.369    Levenberg-Marquardt = 0.0001
Free parameter values:
    22.97      0.8566      28.11      0.07417
Total time = 133.6526s    Reduced Chi^2 = 1.3704    Levenberg-Marquardt = 0.001
Free parameter values:
    22.97      0.8566      28.11      0.07417
Total time = 141.1081s    Reduced Chi^2 = 1.3699    Levenberg-Marquardt = 0.01
Free parameter values:
    22.97      0.8566      28.11      0.07417
Total time = 148.4915s    Reduced Chi^2 = 1.385    Levenberg-Marquardt = 1
Free parameter values:
    22.97      0.8565      28.11      0.0742
Total time = 155.8959s    Reduced Chi^2 = 1.3717    Levenberg-Marquardt = 100
Free parameter values:
    22.97      0.8564      28.12      0.07422
Total time = 163.2728s    Reduced Chi^2 = 1.3693    Levenberg-Marquardt = 10000
Free parameter values:
    22.97      0.8564      28.12      0.07422
*** No improvement in chi-squared over previous iteration ***
Total time = 163.2803s    Reduced Chi^2 = 1.36
Free parameter values:
    22.97      0.8564      28.12      0.07422
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
    1          1
    2          8
    3      22.97 +/- 0.5497
    4      0.8564 +/- 0.01795
    5          0
    6      28.12 +/- 0.1017
    7          0
    8          0
    9          0
   10          0

Background parameter values:
-----
    1      0.07422 +/- 0.003314
    2          0
    3          0

```

Covariance matrix for free parameters:

```
-----
1.0000    0.8639    0.3104   -0.9650
0.8639    1.0000    0.3644   -0.9601
0.3104    0.3644    1.0000   -0.4163
-0.9650   -0.9601   -0.4163    1.0000
-----
```

**** En = 85±5 going to 105

Beginning fit (max 20 iterations)-----

Starting point

Total time = 8.6842s Reduced Chi^2 = 655.9907

Free parameter values:

22.97 0.8564 28.12 0.2478

Iteration = 1

Total time = 43.579s Reduced Chi^2 = 3.8038 Levenberg-Marquardt = 0.1

Free parameter values:

23.23 0.7302 29.09 0.06887

Iteration = 2

Total time = 78.2765s Reduced Chi^2 = 1.2167 Levenberg-Marquardt = 0.01

Free parameter values:

27.97 0.8564 29.72 0.04408

Iteration = 3

Total time = 112.7405s Reduced Chi^2 = 1.1643 Levenberg-Marquardt = 0.001

Free parameter values:

28.81 0.8832 29.82 0.0415

Iteration = 4

Total time = 146.8519s Reduced Chi^2 = 1.1649 Levenberg-Marquardt = 0.0001

Free parameter values:

28.79 0.8829 29.81 0.0415

Total time = 155.4048s Reduced Chi^2 = 1.1642 Levenberg-Marquardt = 0.001

Free parameter values:

28.79 0.8829 29.81 0.0415

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```
-----
1          1
2          8
3      28.79 +/-  0.9787
4      0.8829 +/-  0.02232
5          0
6      29.81 +/-  0.1379
7          0
8          0
9          0
10         0
-----
```

Background parameter values:

```
-----
1      0.0415  +/-  0.002802
2      0
3      0
```

Covariance matrix for free parameters:

```
-----
1.0000  0.9182  0.7234 -0.8036
0.9182  1.0000  0.6792 -0.9025
0.7234  0.6792  1.0000 -0.5396
-0.8036 -0.9025 -0.5396  1.0000
```

**** En = 95±5 going to 105

Beginning fit (max 20 iterations)-----

Starting point

```
-----
Total time = 13.041s   Reduced Chi^2 = 970.3436
Free parameter values:
    28.79      0.8829      29.81      0.2256
```

Iteration = 1

```
-----
Total time = 61.4419s   Reduced Chi^2 = 4.5208   Levenberg-Marquardt = 0.1
Free parameter values:
    25.95      0.7219      30.36      0.05504
```

Iteration = 2

```
-----
Total time = 110.6231s   Reduced Chi^2 = 0.97322   Levenberg-Marquardt = 0.01
Free parameter values:
    30.86      0.8496      30.82      0.03197
```

Iteration = 3

```
-----
Total time = 163.2781s   Reduced Chi^2 = 0.921   Levenberg-Marquardt = 0.001
Free parameter values:
    31.61      0.8732      30.91      0.02978
```

Iteration = 4

```
-----
Total time = 211.5591s   Reduced Chi^2 = 0.92181   Levenberg-Marquardt = 0.0001
Free parameter values:
    31.6      0.8733      30.9      0.02976
Total time = 223.6436s   Reduced Chi^2 = 0.9224   Levenberg-Marquardt = 0.001
Free parameter values:
    31.6      0.8733      30.9      0.02976
Total time = 235.852s   Reduced Chi^2 = 0.91419   Levenberg-Marquardt = 0.01
Free parameter values:
    31.6      0.8733      30.9      0.02976
```

Iteration = 5

```
-----
Total time = 284.1178s   Reduced Chi^2 = 0.91847   Levenberg-Marquardt = 0.001
Free parameter values:
    31.6      0.8732      30.9      0.02976
Total time = 296.2699s   Reduced Chi^2 = 0.92028   Levenberg-Marquardt = 0.01
Free parameter values:
```

```

      31.6      0.8732      30.9      0.02976
Total time = 308.3919s   Reduced Chi^2 = 0.91741   Levenberg-Marquardt = 0.1
Free parameter values:
      31.6      0.8732      30.9      0.02976
Total time = 320.6037s   Reduced Chi^2 = 0.92139   Levenberg-Marquardt = 10
Free parameter values:
      31.6      0.8733      30.9      0.02976
Total time = 332.8284s   Reduced Chi^2 = 0.91426   Levenberg-Marquardt = 1000
Free parameter values:
      31.6      0.8733      30.9      0.02976
Total time = 344.8996s   Reduced Chi^2 = 0.92224   Levenberg-Marquardt = 100000
Free parameter values:
      31.6      0.8733      30.9      0.02976
*** No improvement in chi-squared over previous iteration ***
Total time = 344.907s   Reduced Chi^2 = 0.91419
Free parameter values:
      31.6      0.8733      30.9      0.02976

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	31.6	+/-	0.7663
4	0.8733	+/-	0.01481
5	0		
6	30.9	+/-	0.1029
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.02976	+/-	0.001273
2	0		
3	0		

Covariance matrix for free parameters:

1.0000	0.9262	0.6999	-0.9862
0.9262	1.0000	0.6839	-0.9721
0.6999	0.6839	1.0000	-0.7440
-0.9862	-0.9721	-0.7440	1.0000

**** En = 105±5 going to 105

Beginning fit (max 20 iterations)-----

Starting point

```

Total time = 14.5979s   Reduced Chi^2 = 823.3353
Free parameter values:
      31.6      0.8733      30.9      0.1946

```

Iteration = 1

```

Total time = 71.5362s    Reduced Chi^2 = 4.1152    Levenberg-Marquardt = 0.1
Free parameter values:
    28.22    0.7243    31.48    0.04469
-----
Iteration = 2
-----
Total time = 127.778s    Reduced Chi^2 = 1.1563    Levenberg-Marquardt = 0.01
Free parameter values:
    32.68    0.8362    31.91    0.02673
-----
Iteration = 3
-----
Total time = 184.4802s    Reduced Chi^2 = 1.1336    Levenberg-Marquardt = 0.001
Free parameter values:
    33.2    0.8522    31.97    0.02546
-----
Iteration = 4
-----
Total time = 241.0889s    Reduced Chi^2 = 1.1316    Levenberg-Marquardt = 0.0001
Free parameter values:
    33.2    0.8524    31.97    0.02544
-----
Iteration = 5
-----
Total time = 297.4974s    Reduced Chi^2 = 1.1327    Levenberg-Marquardt = 1e-05
Free parameter values:
    33.2    0.8524    31.97    0.02545
Total time = 311.729s    Reduced Chi^2 = 1.1322    Levenberg-Marquardt = 0.0001
Free parameter values:
    33.2    0.8524    31.97    0.02545
Total time = 325.8718s    Reduced Chi^2 = 1.1344    Levenberg-Marquardt = 0.001
Free parameter values:
    33.2    0.8524    31.97    0.02545
Total time = 341.6208s    Reduced Chi^2 = 1.1284    Levenberg-Marquardt = 0.1
Free parameter values:
    33.2    0.8524    31.97    0.02545
-----
Iteration = 6
-----
Total time = 400.3457s    Reduced Chi^2 = 1.1331    Levenberg-Marquardt = 0.01
Free parameter values:
    33.22    0.8526    31.98    0.02543
Total time = 414.4724s    Reduced Chi^2 = 1.1328    Levenberg-Marquardt = 0.1
Free parameter values:
    33.21    0.8525    31.98    0.02543
Total time = 428.7445s    Reduced Chi^2 = 1.1317    Levenberg-Marquardt = 1
Free parameter values:
    33.21    0.8524    31.97    0.02544
Total time = 443.0894s    Reduced Chi^2 = 1.1346    Levenberg-Marquardt = 100
Free parameter values:
    33.2    0.8524    31.97    0.02545
Total time = 457.4486s    Reduced Chi^2 = 1.1334    Levenberg-Marquardt = 10000
Free parameter values:
    33.2    0.8524    31.97    0.02545
Total time = 471.5155s    Reduced Chi^2 = 1.1314    Levenberg-Marquardt = 1000000
Free parameter values:
    33.2    0.8524    31.97    0.02545
*** No improvement in chi-squared over previous iteration ***
Total time = 471.523s    Reduced Chi^2 = 1.1284
Free parameter values:
    33.2    0.8524    31.97    0.02545
-----
Fit converged:

```

Parameter values (free parameters with error estimates):

Foreground parameter values:

1 1
2 8
3 33.2 +/- 0.8852
4 0.8524 +/- 0.0162
5 0
6 31.97 +/- 0.1227
7 0
8 0
9 0
10 0

Background parameter values:

1 0.02545 +/- 0.001212
2 0
3 0

Covariance matrix for free parameters:

1.0000 0.8941 0.6720 -0.9787
0.8941 1.0000 0.6404 -0.9620
0.6720 0.6404 1.0000 -0.7190
-0.9787 -0.9620 -0.7190 1.0000

Processing cuts: Ei200_off200Gpmiff

```
fit_src_struct = GP_FFCutsFits;  
save(Cutfitfile,"fit_src_struct",-v7.3')
```